Complete Reference Guide

Strings · Arrays · Math · Numbers · Objects · JSON · Dates · Globals

83 Functions · 8 Categories · 100% Built-in

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String Functions

17 functions

.length

Returns number of characters in a string

```
"hello".length // 5
```

Use: Validate input length

.toUpperCase()

Converts all characters to uppercase

```
"hello".toUpperCase() // "HELLO"
```

Use: Normalize text for comparison

.toLowerCase()

Converts all characters to lowercase

```
"HELLO".toLowerCase() // "hello"
```

Use: Case-insensitive matching

.trim()

Removes whitespace from both ends

```
" hi ".trim() // "hi"
```

Use: Clean user form input

.includes()

Returns true if substring is found

```
"hello".includes("ell") // true
```

Use: Check if string contains value

.startsWith()

Returns true if string begins with value

```
"hello".startsWith("he") // true
```

Use: URL or prefix validation

.endsWith()

Returns true if string ends with value

```
"photo.png".endsWith(".png") // true
```

Use: File extension checks

.slice()

Extracts part of string by start/end index

```
"hello".slice(1,3) // "el"
```

Use: Get substrings

.split()

Splits string into array by separator

```
"a,b,c".split(",") // ["a","b","c"]
```

Use: Parse CSV, URLs, sentences

.replace()

Replaces first matching pattern

```
"hi hi".replace("hi","bye") // "bye hi"
```

Use: Simple text substitution

.replaceAll()

Replaces all matching patterns

```
"hi hi".replaceAll("hi","bye") // "bye bye"
```

Use: Global text substitution

.indexOf()

Returns index of first occurrence (-1 if none)

```
"hello".indexOf("l") // 2
```

Use: Find position of substring

.repeat()

Returns string repeated n times

```
"ab".repeat(3) // "ababab"
```

Use: Build separator or padding lines

.padStart()

Pads beginning to reach target length

```
"5".padStart(3,"0") // "005"
```

Use: Format numbers like 01, 02, 03

.padEnd()

Pads end to reach target length

```
"hi".padEnd(5,".") // "hi..."
```

Use: Fixed-width text columns

.charAt()

Returns character at given index

```
"hello".charAt(1) // "e"
```

Use: Read a specific character

.charCodeAt()

Returns Unicode code at given index

```
"A".charCodeAt(0) // 65
```

Use: Encoding or encryption logic

Array Functions

21 functions

.push()

Adds one or more items to end of array

```
let a=[1,2]; a.push(3); // [1,2,3]
```

Use: Append items to a list

.pop()

Removes and returns last item

```
[1,2,3].pop() // 3
```

Use: Stack (LIFO) data structure

.shift()

Removes and returns first item

```
[1,2,3].shift() // 1
```

Use: Queue (FIFO) operations

.unshift()

Adds items to beginning of array

```
let a=[2,3]; a.unshift(1); // [1,2,3]
```

Use: Prepend to a list

.map()

Creates new array by transforming each item

```
[1,2,3].map(x=>x*2) // [2,4,6]
```

Use: Transform data lists

.filter()

Creates new array with items passing a test

```
[1,2,3,4].filter(x=>x>2) // [3,4]
```

Use: Filter search results

.reduce()

Reduces array to a single accumulated value

```
[1,2,3].reduce((a,b)=>a+b,0) // 6
```

Use: Sum totals, combine values

.find()

Returns first item that passes test

```
[1,2,3].find(x=>x>1) // 2
```

Use: Find a user or record by ID

.findIndex()

Returns index of first matching item

```
[1,2,3].findIndex(x=>x>1) // 1
```

Use: Locate item position in list

.includes()

Checks if value exists in array

```
[1,2,3].includes(2) // true
```

Use: Membership / permission checks

.join()

Joins all elements into a string

```
[1,2,3].join("-") // "1-2-3"
```

Use: Build CSV or display strings

.slice()

Returns shallow copy of portion of array

```
[1,2,3,4].slice(1,3) // [2,3]
```

Use: Pagination, copy segments

.splice()

Adds or removes items in place (mutates)

```
let a=[1,2,3]; a.splice(1,1); // [1,3]
```

Use: Edit list in-place

.sort()

Sorts array in place

```
[3,1,2].sort((a,b)=>a-b) // [1,2,3]
```

Use: Sort names, numbers, dates

.reverse()

Reverses array in place

```
[1,2,3].reverse() // [3,2,1]
```

Use: Show newest items first

.flat()

Flattens nested arrays by depth

```
[[1,2],[3]].flat() // [1,2,3]
```

Use: Normalize nested data

.forEach()

Calls a function for each element

```
[1,2].forEach(x=>console.log(x))
```

Use: Logging, DOM updates (side effects)

.every()

Returns true if ALL items pass test

```
[2,4,6].every(x=>x%2===0) // true
```

Use: Form validation checks

.some()

Returns true if ANY item passes test

```
[1,2,3].some(x=>x>2) // true
```

Use: Check permissions list

Array.from()

Creates array from any iterable

```
Array.from("abc") // ["a","b","c"]
```

Use: Convert NodeList or Set to array

Array.isArray()

Returns true if value is an array

```
Array.isArray([1,2]) // true
```

Use: Type checking before operations

Math Functions

12 functions

Math.round()

Rounds to nearest integer

```
Math.round(4.6) // 5
```

Use: Display prices, scores

Math.floor()

Rounds down to nearest integer

```
Math.floor(4.9) // 4
```

Use: Pagination page numbers

Math.ceil()

Rounds up to nearest integer

```
Math.ceil(4.1) // 5
```

Use: Calculate rows needed in a grid

Math.abs()

Returns absolute (positive) value

```
Math.abs(-5) // 5
```

Use: Distance, difference calc

Math.max()

Returns largest value from list

```
Math.max(1,5,3) // 5
```

Use: Find highest score or price

Math.min()

Returns smallest value from list

```
Math.min(1,5,3) // 1
```

Use: Find lowest price

Math.pow()

Raises base to the given exponent

```
Math.pow(2,8) // 256
```

Use: Compound interest, area formulas

Math.sqrt()

Returns the square root of a number

```
Math.sqrt(16) // 4
```

Use: Distance formula, geometry

Math.random()

Returns random float between 0 and 1

```
Math.floor(Math.random()*10) // 0-9
```

Use: Games, shuffle, random picks

Math.trunc()

Removes decimal part without rounding

```
Math.trunc(4.9) // 4
```

Use: Integer division

Math.sign()

Returns 1, -1, or 0 based on sign

```
Math.sign(-5) // -1
```

Use: Determine direction or polarity

Math.PI

Value of pi: 3.14159...

```
Math.PI // 3.14159...
```

Use: Circle area and circumference

Number Functions

8 functions

Number()

Converts a value to number type

```
Number("42") // 42
```

Use: Parse user input strings

.toFixed(n)

Rounds to n decimal places, returns string

```
(3.14159).toFixed(2) // '3.14'
```

Use: Display currency values

.toString(base)

Converts number to string (optional base)

```
(255).toString(16) // "ff"
```

Use: Binary or hex representation

Number.isInteger()

Returns true if value is an integer

```
Number.isInteger(4) // true
```

Use: Input validation

Number.isNaN()

Returns true if value is NaN

```
Number.isNaN(NaN) // true
```

Use: Catch bad parse results

Number.isFinite()

Returns true if value is finite

```
Number.isFinite(Infinity) // false
```

Use: Prevent Infinity in calculations

parseInt()

Parses string and returns integer

```
parseInt("42px") // 42
```

Use: Read integer from text input

parseFloat()

Parses string and returns float

```
parseFloat("3.14abc") // 3.14
```

Use: Read decimal from text input

Object Functions

7 functions

Object.keys()

Returns array of object's own property names

```
Object.keys({a:1,b:2}) // ["a","b"]
```

Use: Iterate over object properties

Object.values()

Returns array of object's own property values

```
Object.values({a:1,b:2}) // [1,2]
```

Use: Sum or display all values

Object.entries()

Returns array of [key, value] pairs

```
Object.entries({a:1}) // [["a",1]]
```

Use: Convert object to iterable list

Object.assign()

Copies properties from sources to target

```
Object.assign({},{a:1},{b:2}) //
{a:1,b:2}
```

Use: Merge config objects

Object.freeze()

Prevents modification of object

```
const o=Object.freeze({a:1})
```

Use: Immutable constants / config

Object.hasOwn()

Returns true if object has own property

```
Object.hasOwn({a:1},"a") // true
```

Use: Safe property existence check

typeof

Returns type of value as string

```
typeof "hi" // "string"
```

Use: Type guards before operations

JSON Functions

2 functions

JSON.stringify()

Converts JS value to JSON string

```
JSON.stringify({a:1}) // '{"a":1}'
```

Use: Send data to API / localStorage

JSON.parse()

Parses JSON string into JS value

```
JSON.parse('{"a":1}') // {a: 1}
```

Use: Read API responses / stored data

Date Functions

6 functions

new Date()

Creates a new Date object

```
new Date() // Thu Apr 23 2026
```

Use: Get current date and time

.getFullYear()

Returns 4-digit year

```
new Date().getFullYear() // 2026
```

Use: Display or compare years

.getMonth()

Returns month index (0-11)

```
new Date().getMonth() // 3 (April)
```

Use: Month-based logic (+1 for display)

.getDate()

Returns day of month (1-31)

```
new Date().getDate() // 23
```

Use: Show current day number

.toLocaleDateString()

Returns locale-formatted date string

```
new Date().toLocaleDateString() //
"4/23/2026"
```

Use: Display readable date to users

Date.now()

Returns current timestamp in ms

```
Date.now() // 1745366400000
```

Use: Measure elapsed time, unique IDs

Global Functions

10 functions

console.log()

Prints output to browser console

```
console.log('hello', 42)
```

Use: Debug during development

setTimeout()

Runs a function after a delay in ms

```
setTimeout(()=>fn(), 1000)
```

Use: Delayed actions, debouncing

setInterval()

Runs a function repeatedly at interval

```
setInterval(()=>tick(), 1000)
```

Use: Clocks, polling, animations

clearTimeout()

Cancels a pending setTimeout

```
const id=setTimeout(fn,1000);
clearTimeout(id)
```

Use: Cancel scheduled actions

clearInterval()

Cancels a running setInterval

```
const id=setInterval(fn,1000);
clearInterval(id)
```

Use: Stop clock or polling loop

fetch()

Makes HTTP request, returns a Promise

```
const res=await fetch('/api'); const
data=await res.json()
```

Use: Call APIs, load data

encodeURIComponent()

Encodes special characters for safe URL

```
encodeURIComponent("a b") // "a%20b"
```

Use: Build query strings safely

decodeURIComponent()

Decodes an encoded URI component

```
decodeURIComponent("a%20b") // "a b"
```

Use: Read URL parameters

structuredClone()

Creates a deep clone of an object

```
const b=structuredClone({a:{c:1}})
```

Use: Copy objects without references

isNaN()

Returns true if value is Not a Number

```
isNaN('hello') // true
```

Use: Quick NaN guard (prefer Number.isNaN)

Quick Cheat Sheet

Most used one-liners

Task	Code
Double all numbers in array	`[1,2,3].map(x => x * 2)`
Filter even numbers	`[1,2,3,4].filter(x => x % 2 === 0)`
Sum all numbers	`[1,2,3].reduce((a,b) => a+b, 0)`
Find item in array	`arr.find(x => x.id === target)`
Remove duplicates	`[...new Set(array)]`
Sort numbers ascending	`arr.sort((a,b) => a - b)`
Round to 2 decimal places	`(3.14159).toFixed(2)`
Random integer 0-9	`Math.floor(Math.random() * 10)`
Check if value is number	`!isNaN(value) && typeof value === 'number'`
Deep copy an object	`structuredClone(obj)`
Convert object to JSON	`JSON.stringify(obj, null, 2)`
Get all object keys as array	`Object.keys(obj)`
Pad number to 2 digits	`String(n).padStart(2, "0")`
Get current year	`new Date().getFullYear()`
Split sentence into words	`"hello world".split(" ")`